Quantum Machine Learning (QML) Classifiers

1. Objective

The primary objective of this project is to design, implement, and evaluate a **hybrid machine learning framework** that integrates **classical machine learning models** with a **Quantum Machine Learning (QML) classifier**.
The project focuses on understanding whether quantum models can learn meaningful patterns from structured data when compared against strong classical baselines.

The key objectives are:

- To preprocess and analyze structured numerical data.
- To build reliable classical machine learning baselines.
- To design and train a **Variational Quantum Classifier (VQC)**.
- To compare classical and quantum models using consistent evaluation metrics.
- To study the feasibility and limitations of QML under current quantum hardware constraints.

2. Overview

This project follows a **multi-stage machine learning workflow**, implemented entirely in a Jupyter notebook environment.

- **Weeks 1–2:**
  Data understanding, preprocessing, exploratory data analysis, feature engineering, and classical machine learning model training.
- **Weeks 3–4:**
  Construction of a complete **Quantum Machine Learning pipeline**, including feature encoding, quantum circuit design, and hybrid training.

The workflow begins with raw numerical data and progresses through scaling, dimensionality reduction, and imbalance handling. Classical models are first trained to establish strong benchmarks. A reduced feature representation is then encoded into a quantum circuit to train a Variational Quantum Classifier.

3. Data Analysis and Preprocessing

3.1 Dataset Description

The dataset used in this project consists of:

- **Numerical input features**
- A **binary target variable** used for classification

The dataset reflects a **class-imbalanced distribution**, which is a common challenge in real-world classification problems.

3.2 Data Cleaning and Imputation

- Missing values were identified across numerical features.
- **Median imputation** was applied to handle missing data.
- Median was chosen due to its robustness against extreme values and outliers.

This step ensured data completeness without losing important samples.

3.3 Feature Scaling

- All numerical features were scaled using **standardization techniques**.
- Feature scaling was essential for:
  - Improving convergence of classical models.
  - Ensuring valid encoding of classical data into quantum circuits.

3.4 Train–Test Split

- The dataset was divided into **training and testing subsets**.
- Stratified splitting was used to maintain class distribution.
- This ensured fair and unbiased evaluation of all models.

3.5 Dimensionality Reduction (PCA)

Due to qubit limitations in quantum computing:

- **Principal Component Analysis (PCA)** was applied.
- A reduced set of the most informative components was retained.
- Classical models were trained on full feature sets, while:
  - **Quantum models used PCA-reduced features only**.

This step enabled efficient quantum encoding while minimizing information loss.

3.6 Handling Class Imbalance

To address class imbalance:

- **Random undersampling** and **SMOTE** techniques were explored.
- SMOTE was applied **only on training data**.
- The test set remained untouched for fair evaluation.

This improved minority-class learning and strengthened classical baselines.

4. Quantum Circuit Design

4.1 Quantum Feature Selection

- A fixed number of PCA components were selected.
- Each selected component was mapped to a **single qubit**.
- This established a one-to-one correspondence between features and qubits.

4.2 Feature Map Construction

- Classical features were encoded using **parameterized quantum rotation gates**.
- Feature values controlled rotation angles.
- Entanglement layers were introduced to capture feature interactions.

This allowed the circuit to model non-linear decision boundaries.

4.3 Variational Ansatz Design

The variational ansatz included:

- Parameterized single-qubit rotation layers.
- Entangling gates between qubits.
- Trainable parameters optimized during learning.

Circuit depth was chosen to balance expressiveness and computational stability.

5. Training Pipeline

5.1 Hybrid Quantum–Classical Learning

The Variational Quantum Classifier follows a hybrid workflow:

1. Classical data is encoded into quantum states.
2. The quantum circuit processes the data.
3. Measurements are extracted from the circuit.
4. A classical optimizer updates circuit parameters.

This loop continues until convergence.

5.2 Quantum Backend

- Training was conducted using a **quantum simulator backend**.
- Finite measurement shots were used to approximate realistic quantum execution.

5.3 Optimization Strategy

- Gradient-free classical optimizers were used.
- These optimizers are suitable for noisy quantum measurement outputs.
- The loss function was based on classification performance.

5.4 Model Evaluation

Models were evaluated using:

- Accuracy
- Classification performance on unseen test data

This ensured consistent comparison between classical and quantum approaches.

6. Classical Machine Learning Model Training and Performance

6.1 Model Selection

The following classical classifiers were trained:

- Logistic Regression
- Random Forest Classifier
- Neural Network (MLP)

These models were selected to provide both linear and non-linear baselines.

6.2 Training Procedure

The classical training pipeline involved:

1. Feature scaling
2. Stratified train–test split
3. Model training on the training set
4. Evaluation on the test set

This consistent process ensured fair benchmarking.

6.3 Performance Comparison

- Ensemble and neural models achieved stronger performance.
- Linear models provided stable but comparatively lower results.
- These results established a solid benchmark for evaluating the quantum classifier.

Accuracy Comparison of Models

Model	Accuracy
Variational Quantum Classifier (VQC)	0.6049
Logistic Regression	0.8393
Random Forest Classifier	0.9476
MLP Classifier (Neural Network)	0.89915

7. Conclusion

This project demonstrates a complete **end-to-end hybrid machine learning system** integrating classical and quantum approaches.

Key conclusions:

- Classical machine learning remains the most reliable solution for structured data today.
- Quantum Machine Learning is feasible on small, reduced datasets.
- Variational Quantum Classifiers can learn meaningful patterns.
- Current hardware and qubit limitations restrict large-scale deployment.

Despite current constraints, this work provides a strong foundation for future exploration into **quantum-enhanced machine learning systems**.